SOLUÇÕES FICHA 2

A. DIFERENCIAÇÃO EM $\mathbb{R}$

A.1. Derivação – revisão e novas aplicações

A.1.1. Regras de derivação

2.1.

a) $-\frac{1}{5}x^{-6/5}+\frac{7x^{4/3}}{3^{2/3}}$

b) $-\frac{5\pi(2x+1)}{(x^2+x+3)^2}$

c) $\frac{3x^6+5x^4}{(x^2+1)^2}+\frac{1}{2\sqrt{x}}$

d) $\frac{3\sqrt{x}}{2}+5x$

e) $e^{-x}(2x-x^2)$

f) $\frac{e^x x-2e^x+x+2}{x^3}$

g) $e^{x^2+\frac{1}{x}+3}\left(2x-\frac{1}{x^2}\right)$

h) $3\ln(x)+3$

i) $\frac{1-\ln(x)}{2x^2}$

j) $3\ln(3x^2+1)+\frac{18x^2}{3x^2+1}$

k) $\frac{\sqrt{x}+3\sqrt{x}\ln(x)}{2[1+\ln(x)]^2}$

l) $\frac{1}{(x+1)(\ln(x+1))}$

m) $\frac{1}{(x+1)(x+2)}$

n) $\frac{e^{1/x}}{x}\left(-\frac{\ln(x)}{x}+1\right)$

o) $\frac{2x[\ln(x^4+x+1)](x^4+x+1)-[\ln(x^2+1)](4x^3+1)(x^2+1)}{[\ln(x^4+x+1)]^2(x^4+x+1)(x^2+1)}$

p) $\frac{e^x[x^2\ln(x+1)-x\ln(x+1)-2\ln(x+1)+x]}{(x+1)\ln(10)\,x^3}$

q) $\frac{1-x\ln(2)\ln(x)}{2^x x\ln(10)}$

r) $ex^{e-1}+e^x+5^{x-1}\ln(5)$

s) $\frac{5\sin(1/x)}{x^2}$

t) $4x^3\cos(x)-x^4\sin(x)$

u) $\sin\left(\frac{\pi}{2}x\right)\sin\left(\frac{1}{x}\right)+x\left[\frac{\pi}{2}\cos\left(\frac{\pi}{2}x\right)\sin\left(\frac{1}{x}\right)-\frac{\sin\left(\frac{\pi}{2}x\right)\cos\left(\frac{1}{x}\right)}{x^2}\right]$

v) $e^x(1+x)\cos(xe^x)$

w) $2x\sec^2(x^2+1)$

x) $\frac{\tan(x^2)}{x}+2x\ln(x)\sec^2(x^2)$

y) $\sec(x)\tan(x)+\frac{1}{x\ln(10)}$

z) $\sec(x)\tan(x) - \operatorname{cosec}(x)\operatorname{cotg}(x)$

aa) $2x\tan(x^3) + 3x^4\sec^2(x^3)$

ab) $-\dfrac{\tan(x)}{\ln(10)} + \dfrac{\sec^2[\log(x)]}{x\ln(10)}$

ac) $\sin\big(2(x+1)\big) + 3x^2\sec^2(x^3)$

ad) $-\dfrac{\sin(\sqrt{x})}{2\sqrt{x}} + \sin(2x)$

ae) $2e^x\sec^2(e^x) - 5\sec(x)\tan(x)$

af) $\dfrac{\cos^2(x)[\sec^2(x) + 2x\cos(x^2+1)] + \sin(2x)[\tan(x) + \sin(x^2+1)]}{\cos^4(x)}$

ag) $\sin(x)^{\tan(x)}[\sec^2(x)\ln(\sin(x)) + 1]$

ah) $\sin(x)^{\tan(x)}[\sec^2(x)\ln(\sin(x)) + 1]$

ai) $\ln(x)^{1/x}\left[-\dfrac{\ln(\ln(x))}{x^2} + \dfrac{1}{x^2\ln(x)}\right]$

aj) $[\sec(x)+3]^{\ln(x)}\left[\dfrac{1}{x}\ln(\sec(x)+3) + \dfrac{\ln(x)\sec(x)\tan(x)}{\sec(x)+3}\right]$

A.1.2. Derivação da função inversa

2.2.

a) $\dfrac{1}{\sqrt{1-x^2}}$

b) $-\dfrac{1}{\sqrt{1-x^2}}$

c) $\dfrac{1}{x^2+1}$

d) $\dfrac{1}{x\sqrt{x^2-1}}$

e) $-\dfrac{1}{x^2+1}$

f) $-\dfrac{1}{x\sqrt{x^2-1}}$

g) $-\dfrac{2}{(1+x^2)\sqrt{x^2+2}}$

h) $\dfrac{x - 2(1+x^2)\arctan(x)}{x^3(1+x^2)}$

i) $7x^6\left(\dfrac{\pi}{2} - \arctan(x)\right) - \dfrac{x^7}{1+x^2}$

j) $\dfrac{\left[\cos\left(\frac{1}{x}\right) - 1\right] - \sin\left(\frac{1}{x}\right)\left[\arctan\left(\frac{1}{x}\right) - \frac{1}{x}\right](x^2+1)}{x^2(x^2+1)\left[\cos\left(\frac{1}{x}\right) - 1\right]^2}$

k) $\dfrac{1}{\arctan\left(\frac{x}{x+1}\right)\ln(10)(2x^2+2x+1)}$

l) $\dfrac{1}{x\ln(x)\sqrt{(\ln^2(x)-1)}}$

m) $-\dfrac{2}{x\sqrt{x^4-1}} + \cos(\sin(\sin(x)))\cos(\sin(x))\cos(x)$

n) $3\,[\arctan(\pi\sqrt{x})]^2\dfrac{\pi}{2\sqrt{x}(1+\pi^2x)}$

A.1.3. Gráficos de funções

2.3. Confirmar os esboços com uma calculadora gráfica.

A.1.4. Derivada da função composta. Regra da cadeia

2.4. $0.25\ cm/min$

2.5. $20\pi\ m^2/s$

2.6. $\frac{8}{\pi r^2}\ cm/s$

2.7. $-960\ m/s$

2.8. $3.75\ m/s$

2.9. $\frac{1}{3}\ m/s$

2.10. $5.33 \times 10^{-3}\ cm/s$

2.11. $\frac{d^2y}{dx^2} = \frac{dy}{du}\frac{d^2u}{dx^2} + \left(\frac{du}{dx}\right)^2 \frac{d^2y}{du^2}$

2.12. $\frac{dy}{dx} = \frac{dy}{du}\frac{du}{dv}\frac{dv}{dx}$

2.13. $f'(x) = \frac{1}{(x+1)^2};\ g'(x) = \frac{1}{(2x+1)^2}$

A.2. Teorema de Cauchy e regra de L'Hôpital

2.14.

a) 0

b) 0

c) e^2

d) 1

e) ∞

f) $1/2$

g) $2/3$

h) e^2

i) $-2/\pi$

A.3. Aproximação polinomial

2.15.

a) $f(x) = \left(\sum_{k=0}^{n} \frac{x^k}{k!}\right) + \left(\frac{e^z}{(n+1)!}x^{n+1}\right), x < z < 0 \ \vee\ 0 < z < x$

b) $f(x) = \left(\sum_{k=0}^{n} (-1)^k \frac{x^{2k}}{(2k)!}\right) + \left((-1)^{n+1}\frac{\cos(z)}{(2n+2)!}x^{2n+2}\right), x < z < 0 \vee\ 0 < z < x$

c) $f(x) = \left(\sum_{k=0}^{n} (-1)^k \frac{(x-\pi/2)^{2k}}{(2k)!}\right) + \left((-1)^{n+1}\frac{\operatorname{sen}(z)}{(2n+2)!}(x-\pi/2)^{2n+2}\right),$

$x < z < \pi/2 \ \vee\ \pi/2 < z < x$

d) $f(x) = \left(\log(2) + \sum_{k=1}^{n}(-1)^{k+1}\frac{(n-2)^k}{(2^k)k}\right) + \left(\frac{(-1)^{n+2}!}{z^{n+1}}\frac{(x-2)^{n+1}}{(x+1)!}\right)$,

$x < z < 2 \vee 2 < z < x$

e) $f(x) = \left(\sum_{k=0}^{n}(-1)^k\frac{(x-2)^k}{2^{k+1}}\right) + \left(\frac{(-1)^{n+1}}{z^{n+2}}(x-2)^{n+1}\right), x < z < 2 \vee 2 < z < x$

f) $f(x) = \left(1 + \sum_{k=1}^{n}(-1)^k\frac{1\times 4\times 7\times \ldots\times(3n-2)}{3^k}\frac{(x-1)^k}{k!}\right) + \left((-1)^{n+1}\frac{1\times 4\times 7\times \ldots\times(3n+1)}{3^{n+1}}z^{-3(n+4)/3}\right)$,

$x < z < 1 \vee 1 < z < x$

A.4. Séries de Taylor com limite dos polinómios de Taylor

2.16.

a) $f(x) = \sum_{n=0}^{\infty}\frac{x^n}{n!}$

b) $f(x) = \sum_{n=0}^{\infty}(-1)^n\frac{x^{2n}}{(2n)!}$

c) $f(x) = \sum_{n=0}^{\infty}(-1)^n\frac{(x-\pi/2)^{2n}}{(2n)!}$

d) $f(x) = \log(2) + \sum_{n=1}^{\infty}(-1)^{n+1}\frac{(x-2)^n}{(2^n)n}$

e) $f(x) = \sum_{n=0}^{\infty}(-1)^n\frac{(n-2)^n}{2^{n+1}}$

f) $f(x) = 1 + \sum_{k=1}^{\infty}(-1)^k\frac{1\times 4\times 7\times \ldots\times(3n-2)}{3^k}\frac{(x-1)^k}{k!}$

2.17.

a) $e^{-6x} = \sum_{n=0}^{\infty} \frac{(-6)^n e^{24}(x+4)^n}{n!}$

b) $\ln(3+4x) = \ln(3) + \sum_{n=1}^{\infty} \frac{(-1)^{n+1}(4/3)^n x^n}{n}$

c) $\frac{7}{x^4} = \sum_{n=0}^{\infty} \frac{7(n+3)(n+2)(n+1)}{6 \times 3^{n+4}}(x+3)^n$

d) $7x^2 - 6x + 1 = 17 + 22(x-2) + 7(x-2)^2$

A.5. Séries numéricas

2.18.

a) $\sum_{n=1}^{\infty} \frac{1}{(2n-1)(2n+1)} = \frac{1}{2}$

b) $\sum_{n=1}^{\infty} \frac{2}{3^{n-1}} = 3$

c) $\sum_{n=1}^{\infty} \frac{1}{n^2-1} = \frac{3}{4}$

d) $\sum_{n=1}^{\infty} \frac{2^n + 3^n}{6^n} = \frac{3}{2}$

e) $\sum_{n=1}^{\infty} \frac{\sqrt{n+1} - \sqrt{n}}{\sqrt{n^2+n}} = 1$

f) $\sum_{n=1}^{\infty} \frac{2n+1}{n^2(n+1)^2} = 1$

2.19.

a) Série Telescópica:
Convergente;
Soma igual a 1.

b) 5 Termos de Série Harmónica e Série Telescópica.
Soma igual a $\frac{147}{60}$

c) Série Divergente

2.20.

a) Série Convergente
$S = \frac{3}{2}$

b) Série Convergente
$S = \frac{1}{2}$

c) Série Divergente

d) Série Divergente

e) Série Divergente

f) Série Convergente
$S = \frac{1}{7} - 1$

2.21.

a) Série Convergente
b) Série Divergente
c) Série Absolut. Convergente
d) Série Absolt. Convergente
e) Série Divergente
f) Série Convergente
g) Série Divergente
h) Série Simpl. Convergente
i) Série Divergente
j) Série Absolut. Convergente
k) Série Absolut. Convergente
l) Série Divergente
m) Série Absolut. Convergente
n) Série Absolut. Convergente